

SE LAB 2

Agile Backlog Creation and Sprint Simulation in Jira

Project: Personalized Meal & Diet Subscription Manager

Name	RUSHIKESH PATIL
SRN	PES1UG24AM394
Jira Space	SE_LAB_JIRA

Software Engineering Lab - Practical Report

1. Objective

The aim of this lab is to create and manage an Agile backlog in Jira and use it to simulate sprint work. I used the Personalized Meal & Diet Subscription Manager scenario and created work items for the main functions of the system.

What I completed in Jira

- Created the project backlog with an Epic and related work items.
- Added priorities to the work items.
- Used Scrum sprints to organise the work.
- Moved work through the Jira board and completed the sprint items.
- Checked the Jira reports, including the burndown and other report views.

Project functions used for the backlog

The backlog covers meal-plan generation, dietary-profile management, dietitian approval, subscription scheduling, delivery tracking, schedule changes and protection of subscriber information.

2. Epic and Agile Backlog

The main Epic in my Jira space is Personalized Meal & Diet Subscription Manager. The work items visible in Jira are listed below.

Key	Type	Priority	Summary
SCRUM-1	Task	High	Manage Dietary Profile
SCRUM-10	Story	High	Protect Subscriber Information
SCRUM-11	Task	Medium	Test Meal Plan Generation
SCRUM-12	Task	Medium	Verify Subscription Schedule
SCRUM-2	Story	High	Generate Weekly Meal Plan
SCRUM-4	Subtask	—	Subtask 2.1
SCRUM-5	Epic	Medium	Personalized Meal & Diet Subscription Manager
SCRUM-6	Story	High	Review and Approve Meal Plan
SCRUM-7	Story	Medium	Manage Subscription Schedule

The table uses the same work-item names, types and priorities shown in my Jira workspace.

3. Jira Backlog / Work-Item Details

This screenshot shows the work-item details from my Jira space. It includes the Epic, Stories, Tasks and Subtask used for the project.

Spaces

SE_LAB_JIRA

Summary Backlog Board Development Timeline Docs Forms Reports

Search table

Work item key	Work type	Work item priority	Issue summary
SCRUM-1	Task	High	Manage Dietary Profile
SCRUM-10	Story	High	Protect Subscriber Information
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SCRUM-12	Task	Medium	Verify Subscription Schedule
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SCRUM-6	Story	High	Review and Approve Meal Plan
SCRUM-7	Story	Medium	Manage Subscription Schedule

Figure 1. Jira Work Item Details

I used this page to check the backlog items before working with the sprints.

4. Sprint Board

I used the Scrum board to organise the work. The board shows the normal flow from To Do to In Progress, In Review and Done.

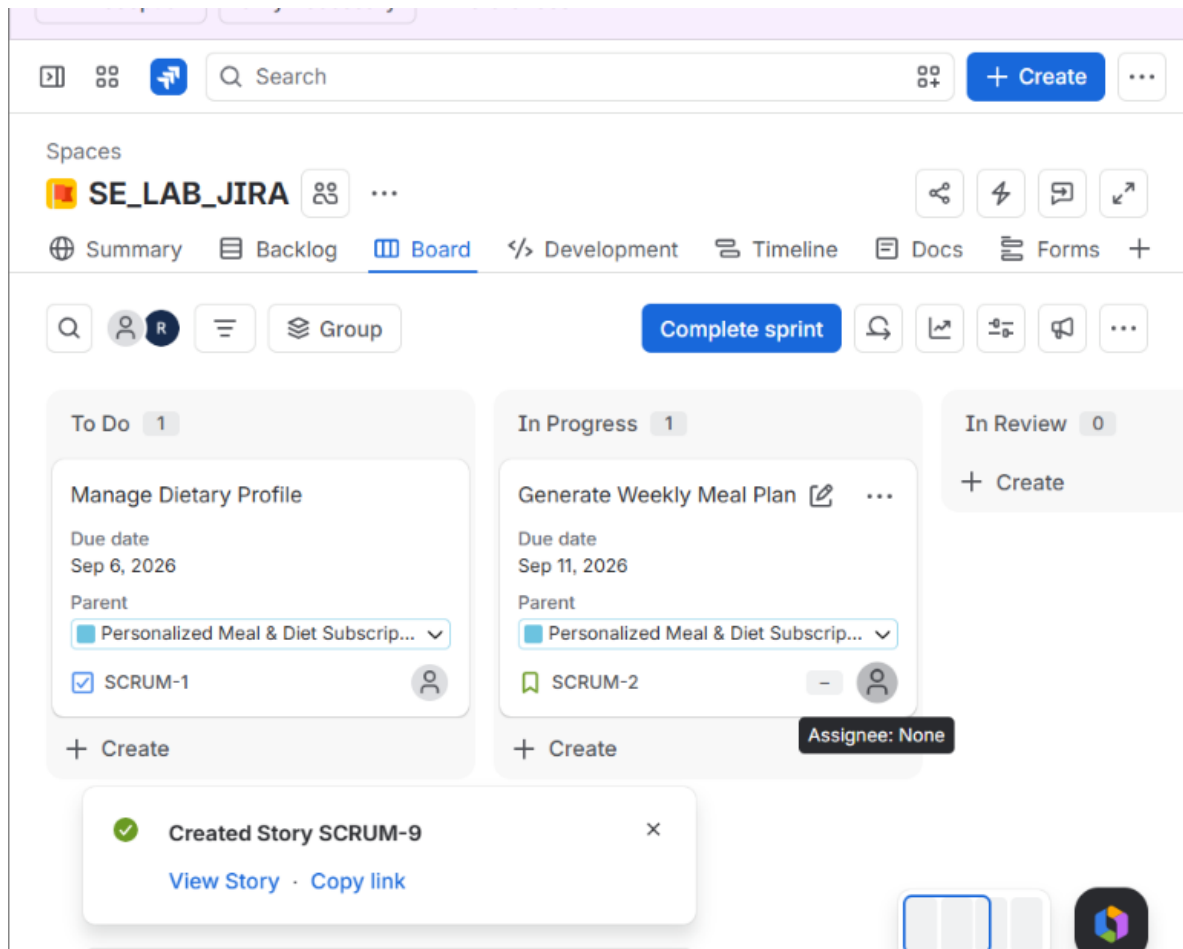


Figure 2. Jira Scrum board

The board was used to move the work items during the sprint simulation.

5. Sprint Completion

After completing the selected work, the items were shown in the Done column. I also created and used another sprint for the remaining work.

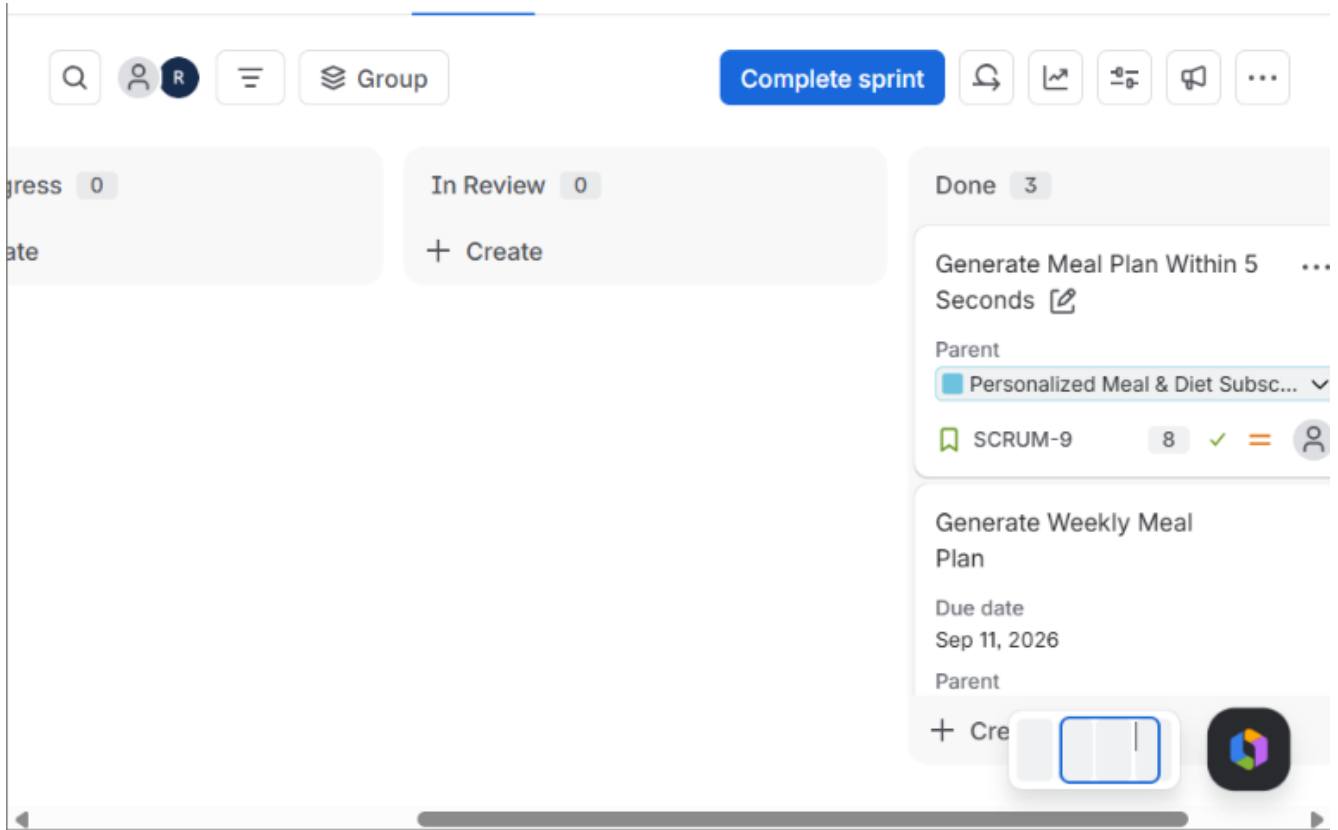


Figure 3. Completed work in Jira

The sprint work was completed and the Jira board was updated.

6. Sprint Burndown

I checked the Sprint Burndown Chart to see how the story points changed during the sprint. The chart is based on the work items and their story-point values in Jira.

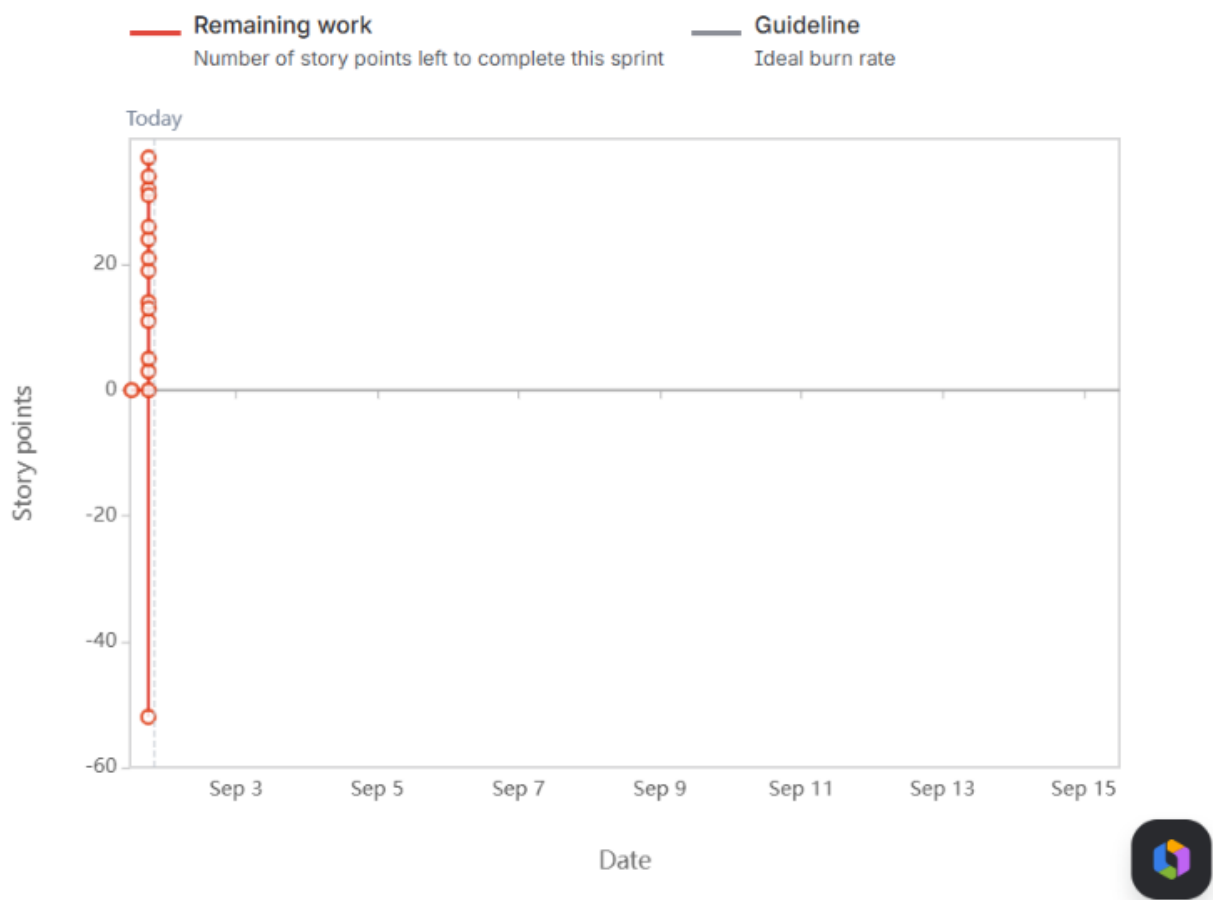


Figure 4. Jira Sprint Burndown Chart

The chart shows the work points at the start of the sprint and the remaining work as the sprint progresses. In this simulation, the work was completed in a short period rather than evenly across the full sprint.

7. Jira Reports

I opened the Reports section in the Jira space to check the project statistics. The reports showed the work-item status, work-item types, assignees and sprint-related information.

Reports checked

- Work items by status
- Work items by type
- Work items by assignee
- Work item creation trend
- Work item cycle time
- Sprint burndown
- Sprint velocity / commitment information

Some Jira report pages need development or deployment data from connected tools. I kept those reports as shown in Jira instead of adding values that were not available.

8. Reflection

1. Did my estimations reflect the actual effort?

The story points gave me a simple way to compare the size of the work items. Since this was a lab simulation, the time taken was not the same as a real software project.

2. Was my backlog well prioritised?

Yes. I kept important functions such as dietary-profile management, meal-plan generation, subscriber-information protection and meal-plan approval at higher priority. Other supporting tasks were kept at medium priority.

3. How did the sprint match my plan?

I added the work to the sprint, used the Scrum board to track it and moved the completed items to Done. I also used another sprint for the remaining work.

4. What did I learn from the burndown chart?

The chart helped me see how much work was left during the sprint. In my simulation, most of the work was completed close to the beginning, so the progress was not spread equally over all the sprint days.

9. Conclusion

This lab helped me understand how a software project can be organised using Jira and Scrum. I created the backlog, arranged the work items, used sprints, tracked progress on the board and checked the reports. The screenshots in this report show the work completed in the SE_LAB_JIRA space.

Final checklist

Lab activity	Status
Epic and backlog created	Completed
Priorities added	Completed
Sprint work organised	Completed
Work moved through the Scrum board	Completed
Sprint work completed	Completed
Burndown checked	Completed
Jira reports checked	Completed

— End of Report —